



A B M **& Digital Twin Boot Camp**



Morphotect



• **Sepehr Doroudi**



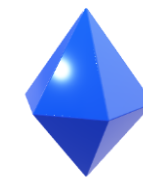
• **Foroozan Almasi**



• **Saba Sanati**

Problem description

Retrofitting the forgotten open space of the University of Science and Technology of Iran



How should we design it based on users' needs?

WHY

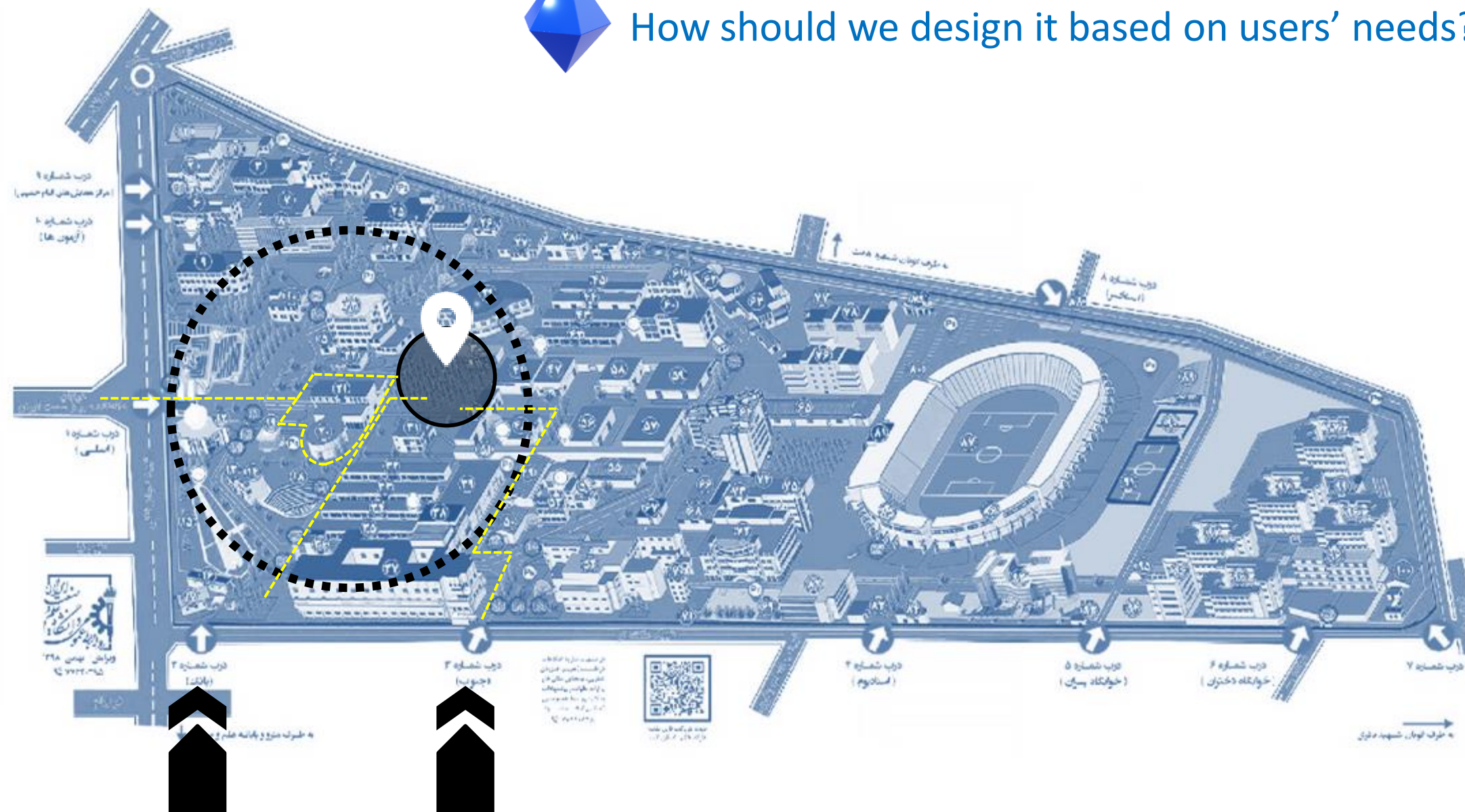
High potential of the open space

Spatial Incomprehensibility.

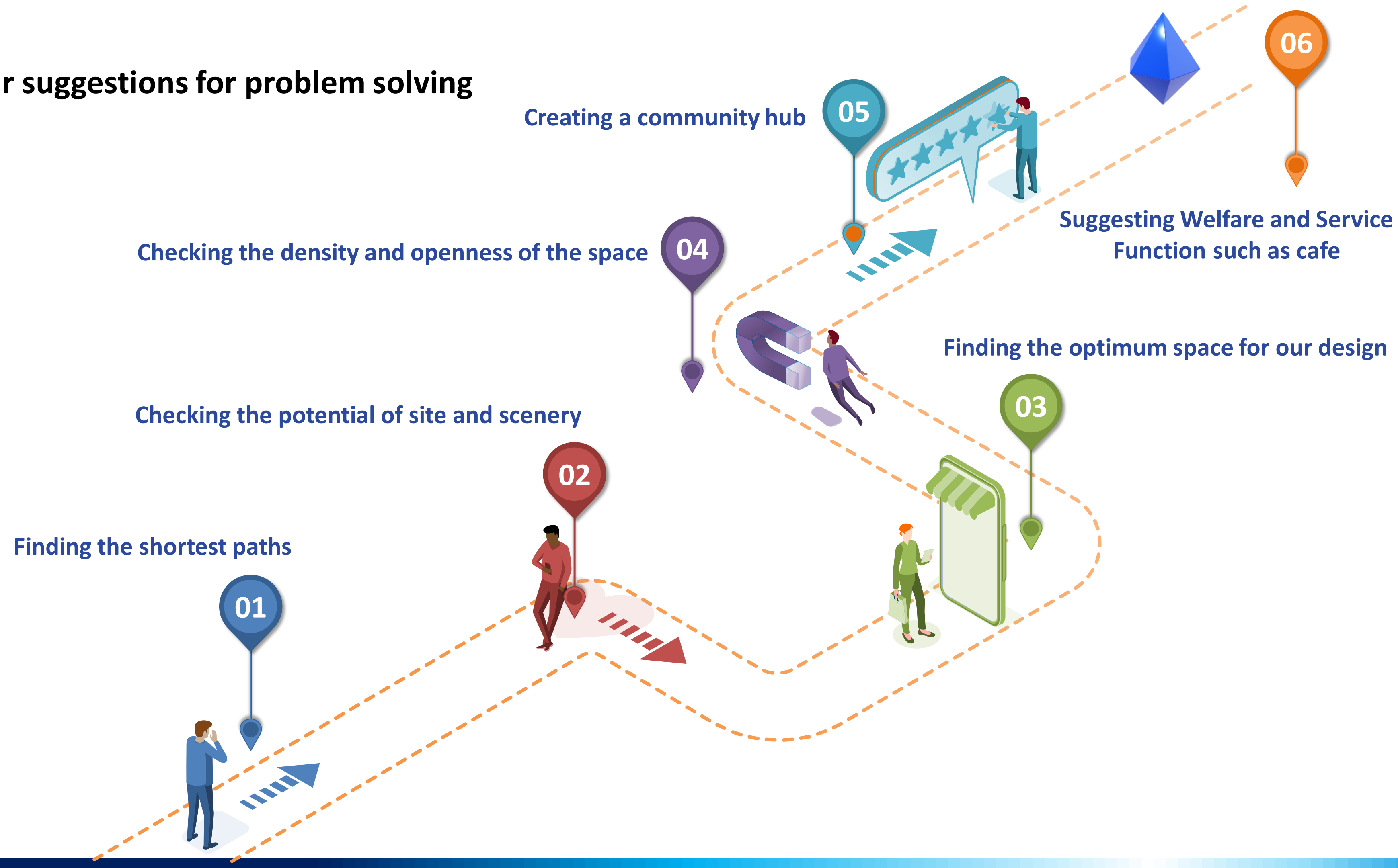
Confusion in Wayfinding

No sense of Place

Lack of proper special organization



Our suggestions for problem solving



Algorithm logic and explanation of code structure



Determining The Target & Starting Points



Modeling The site & Area, Identifying Obstacles In Movements



Find The Shortest Path Between The Target & Starting Points With A* Algorithm



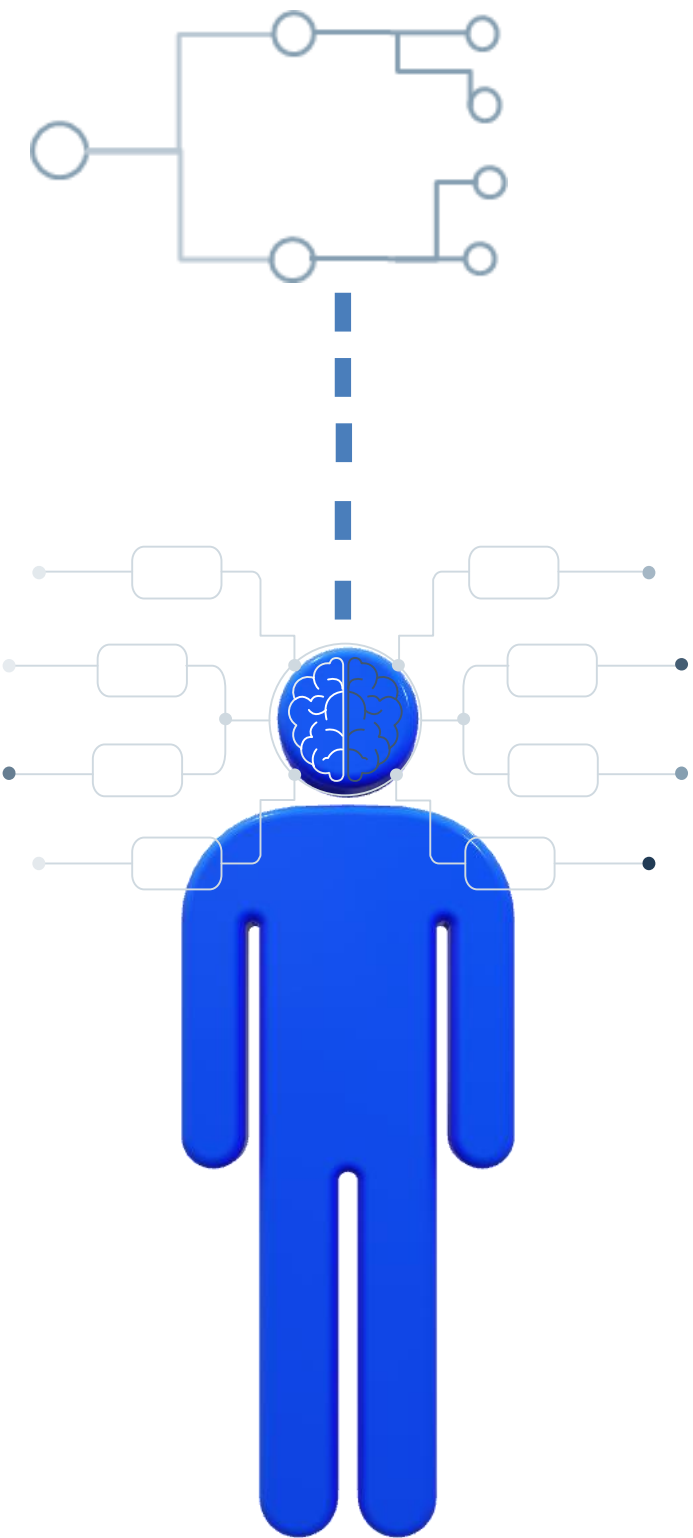
Navigating The Model Along The Specified Path While Identifying Potential Obstacles



Counting The Obstacles Observed On Each Path and determine the openness



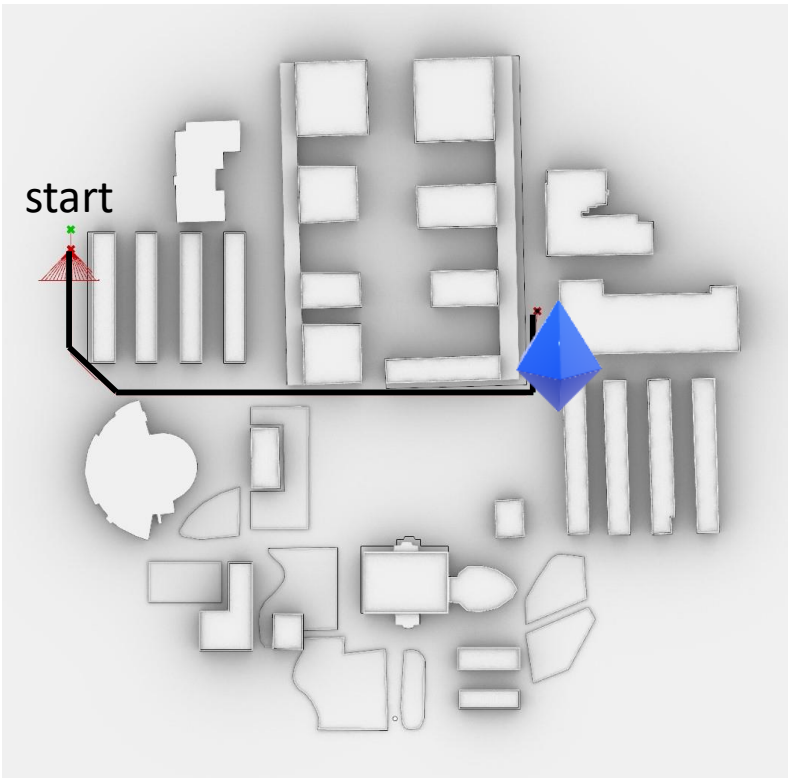
Choosing The Main Path and the main area for public space



Displaying the outputs and the reasons for choosing A*



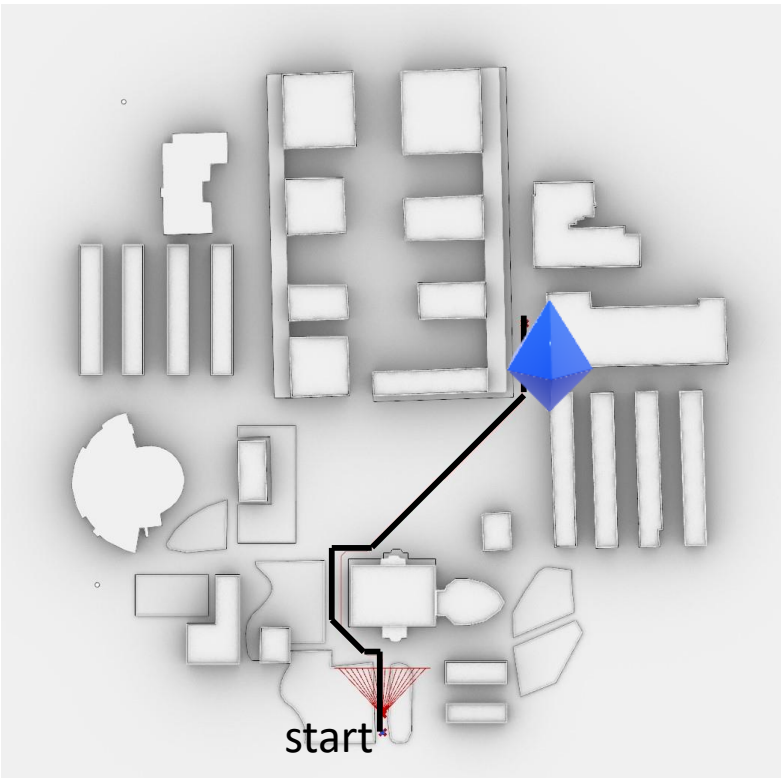
A



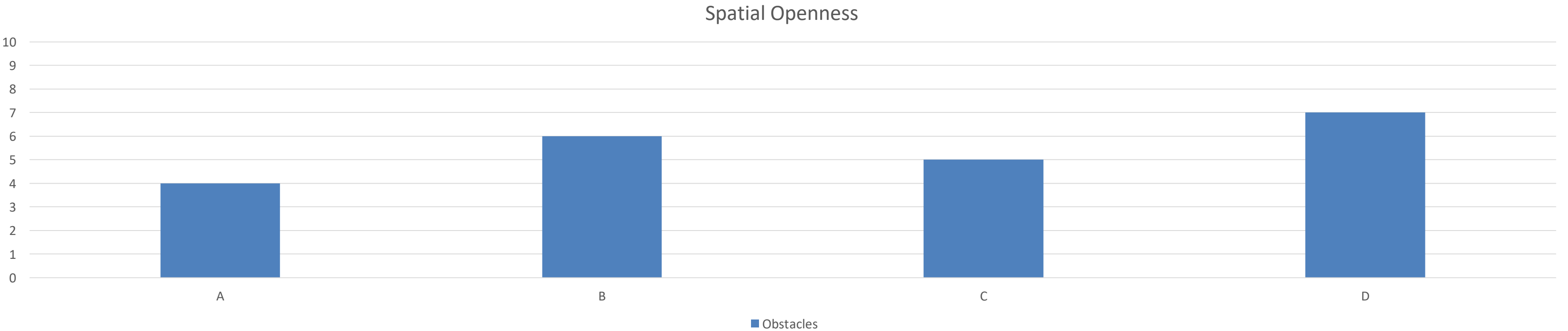
B



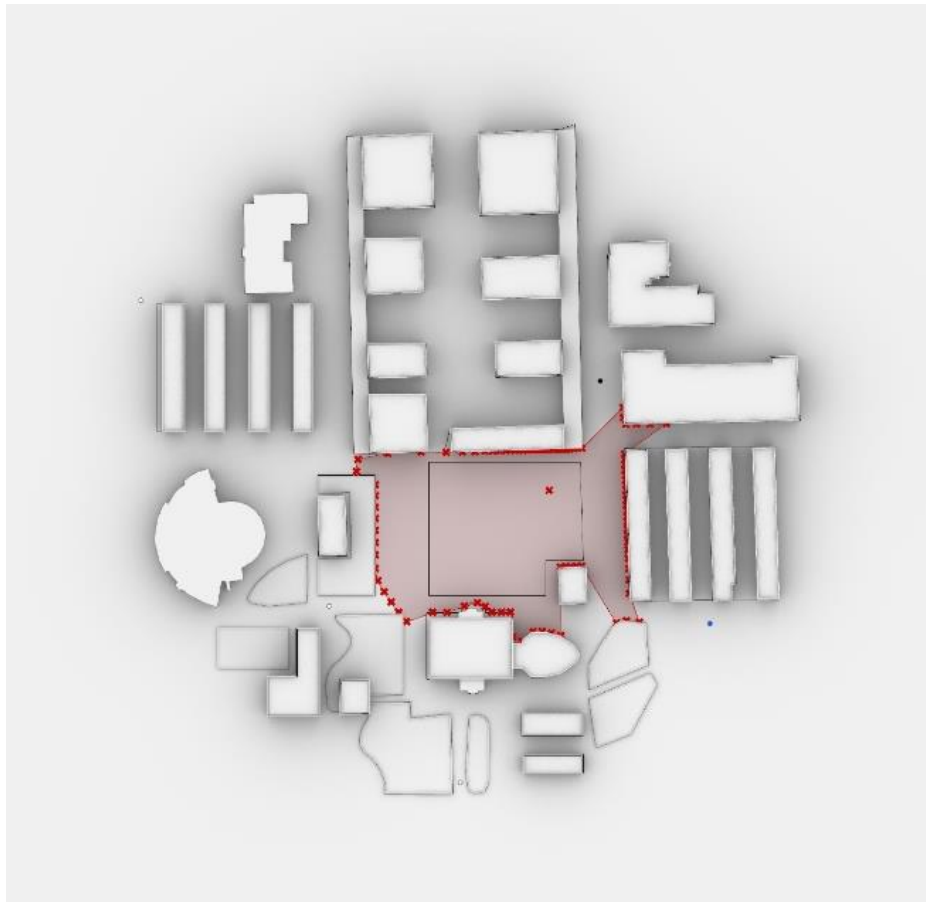
C



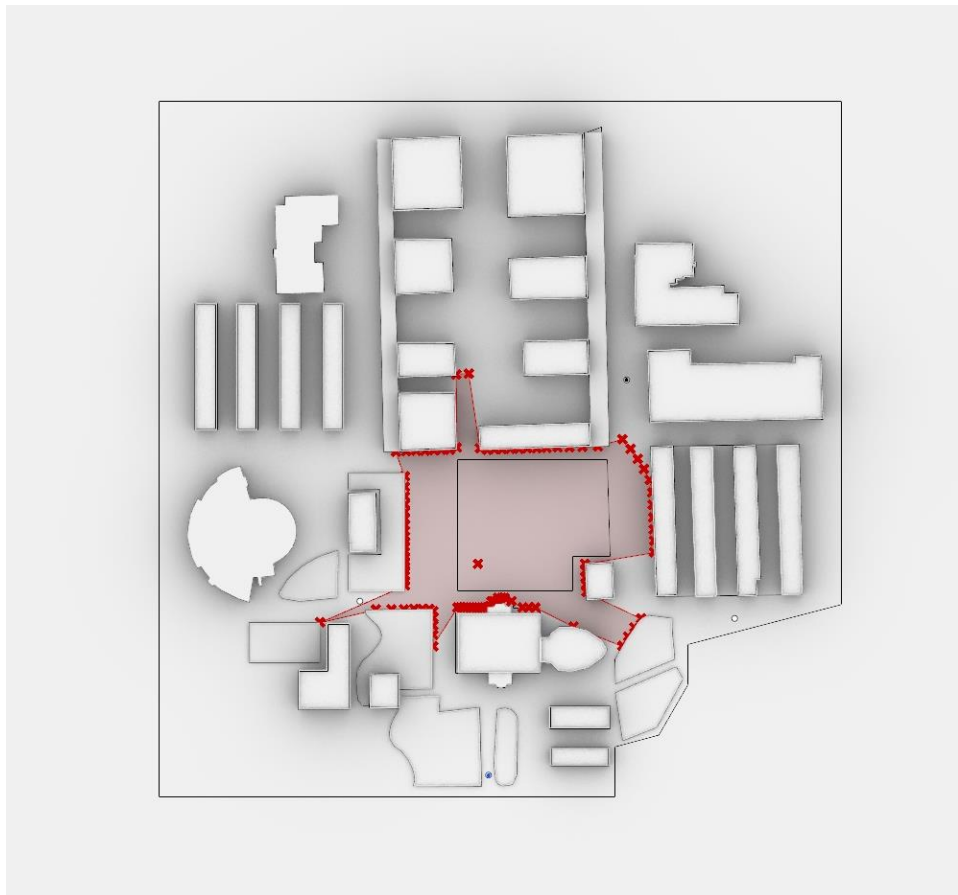
D



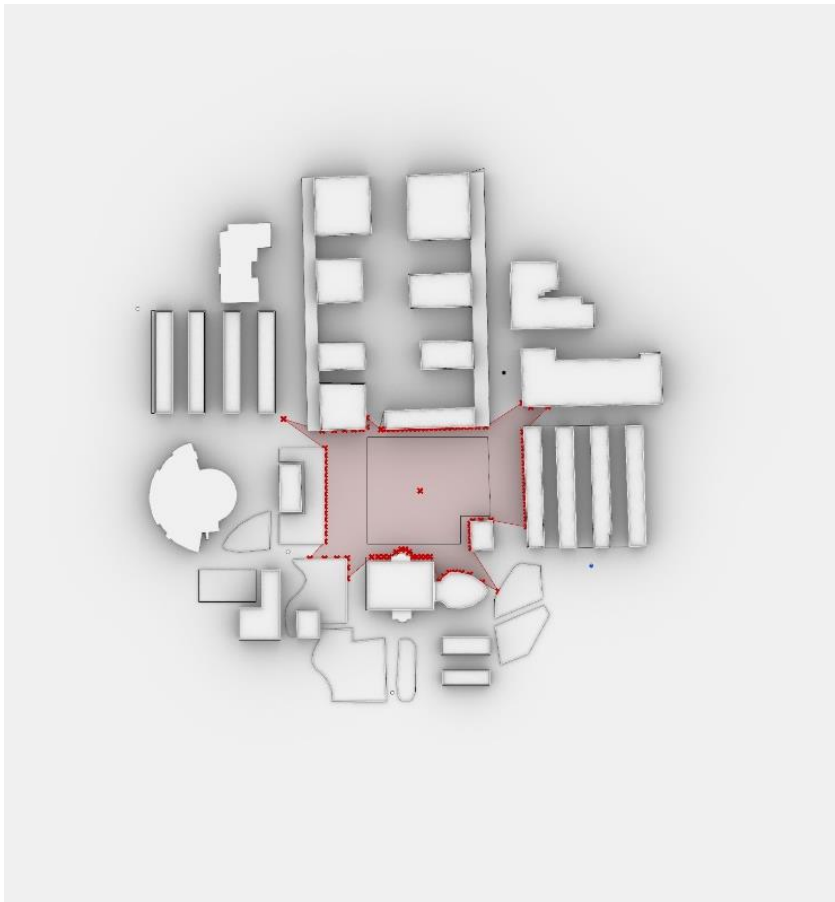
Displaying the outputs and the reasons for choosing Isovist



A

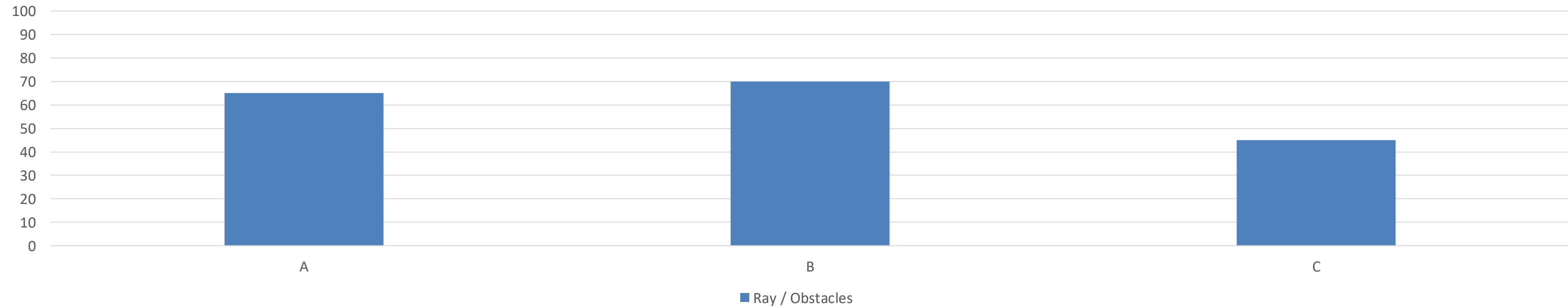


B

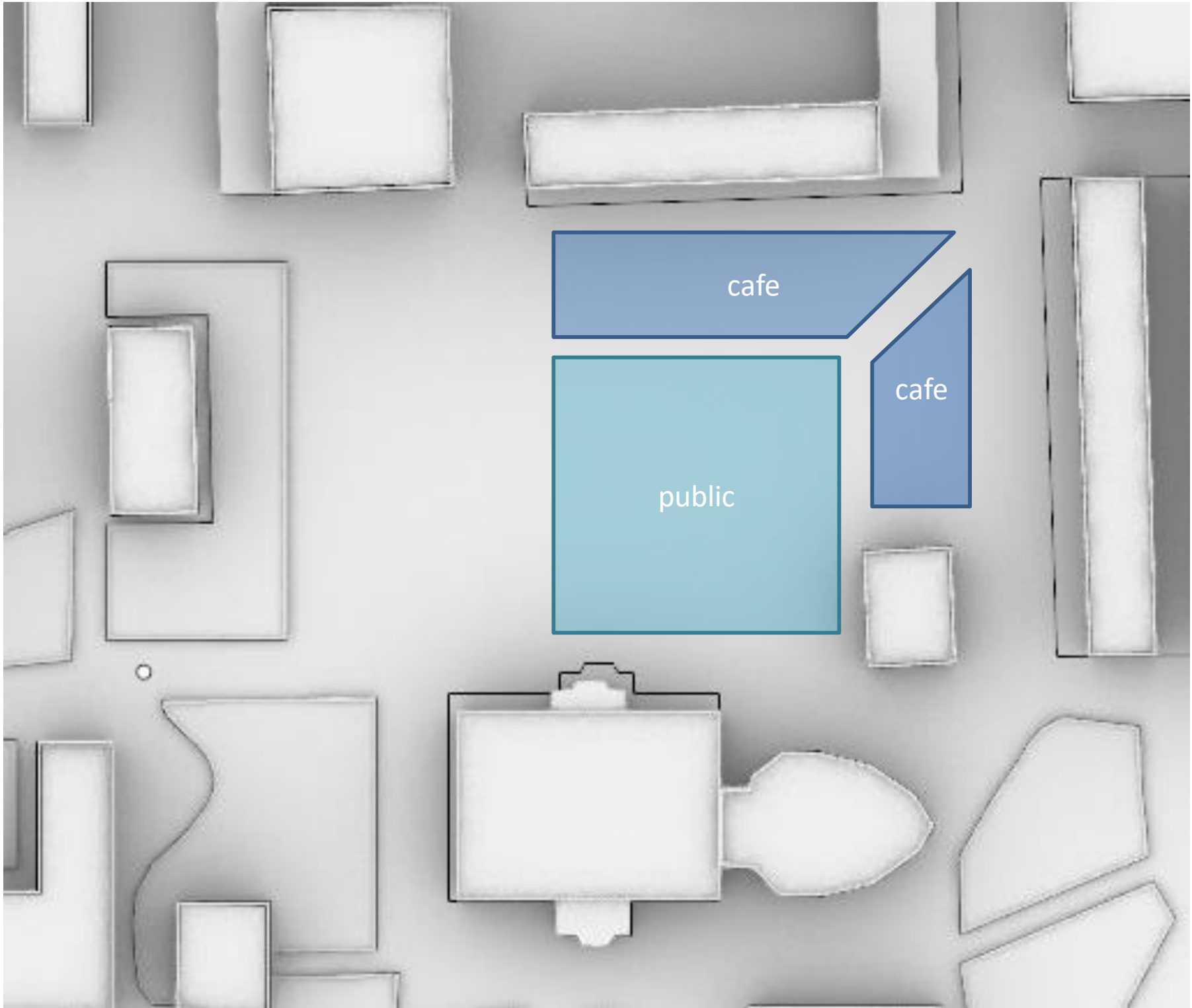
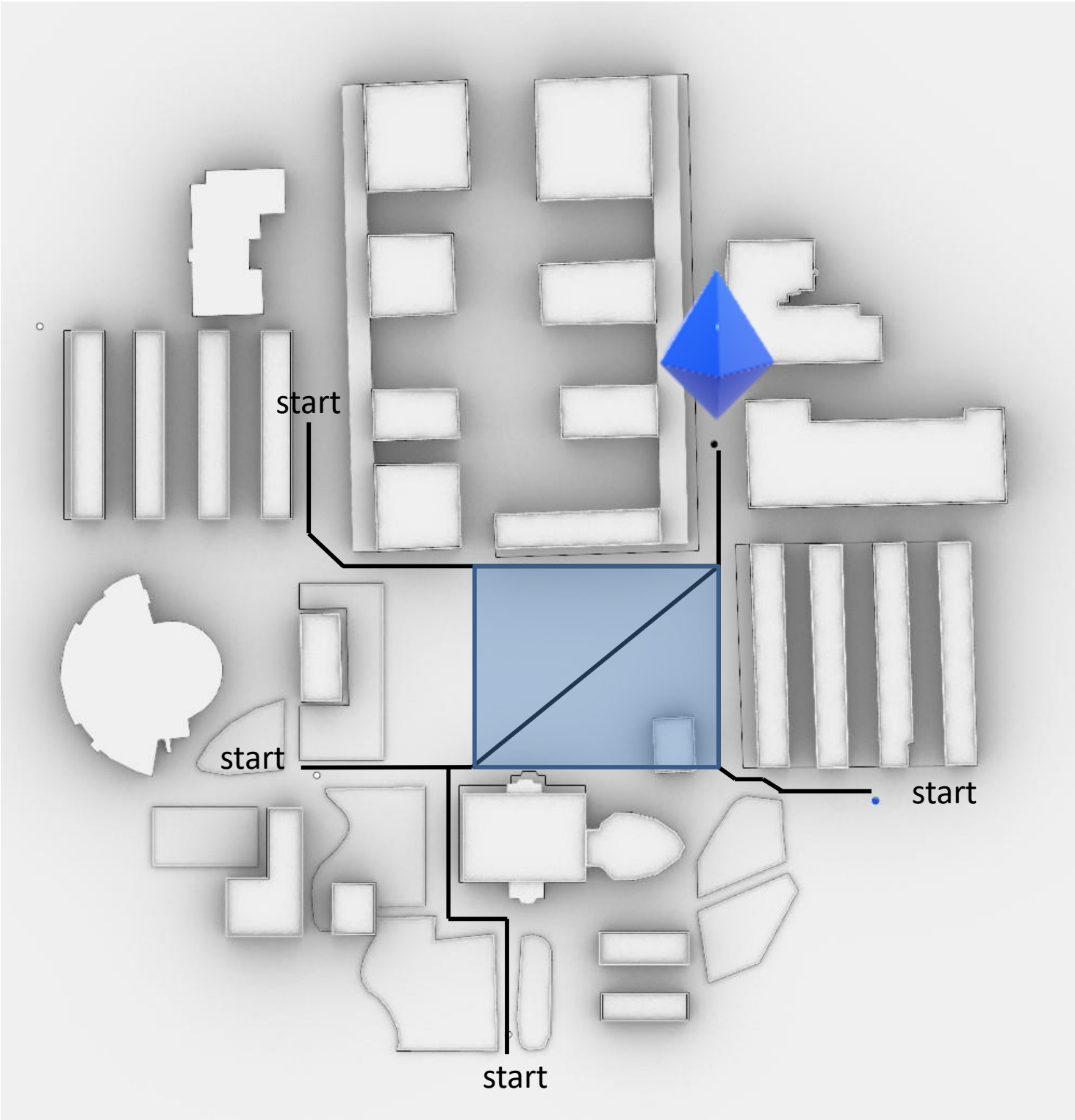


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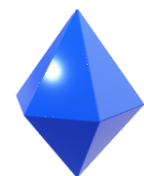
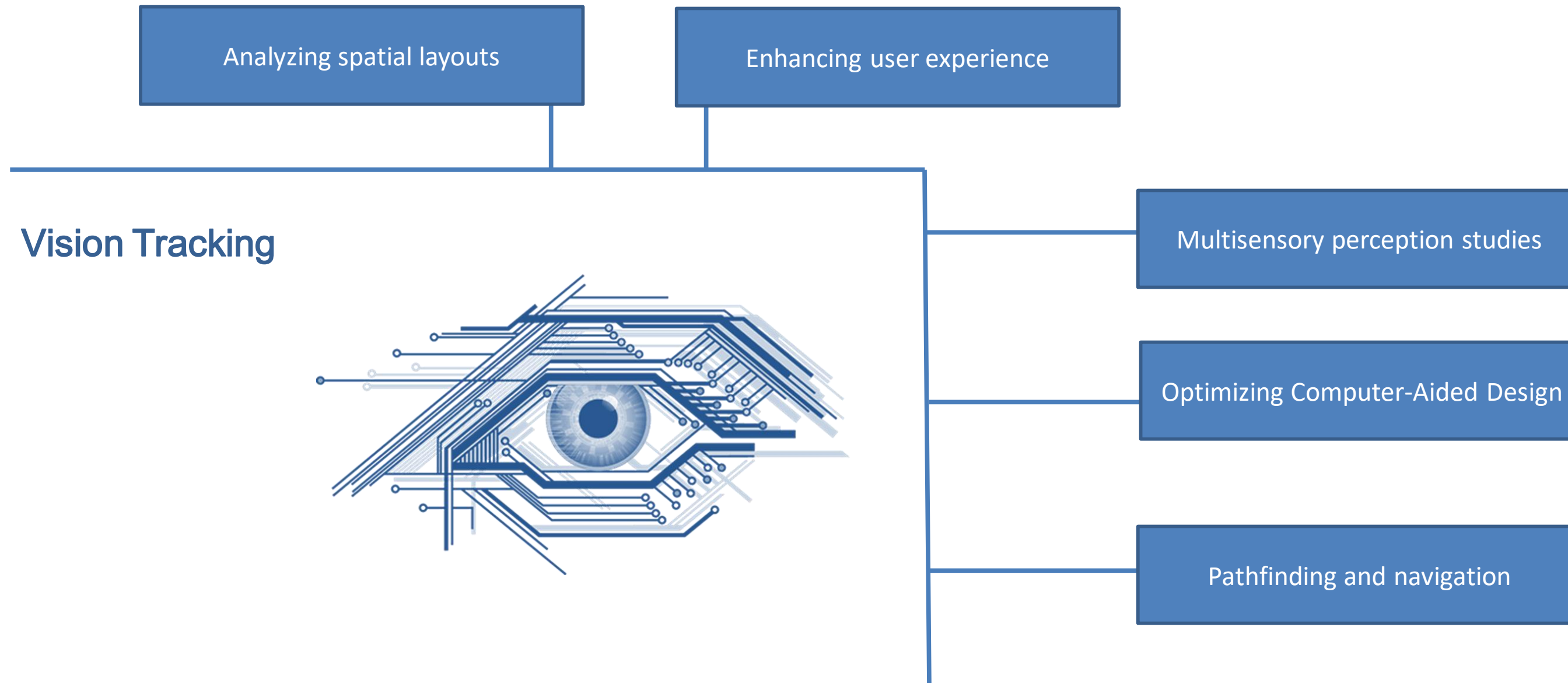
Visual Density



Displaying the outputs and the reasons for choosing these methods

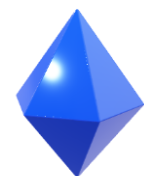
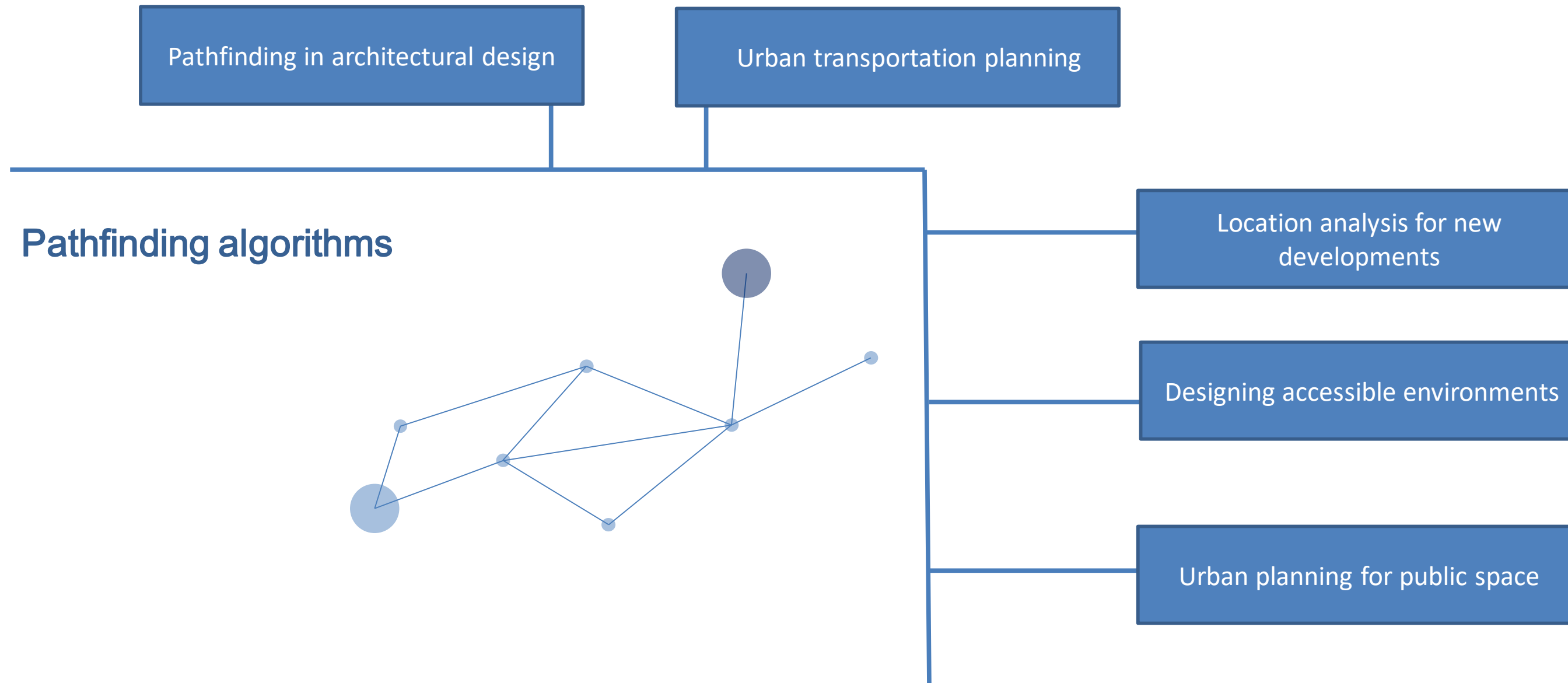


Algorithm usage in architecture and urban planning



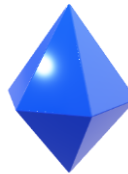
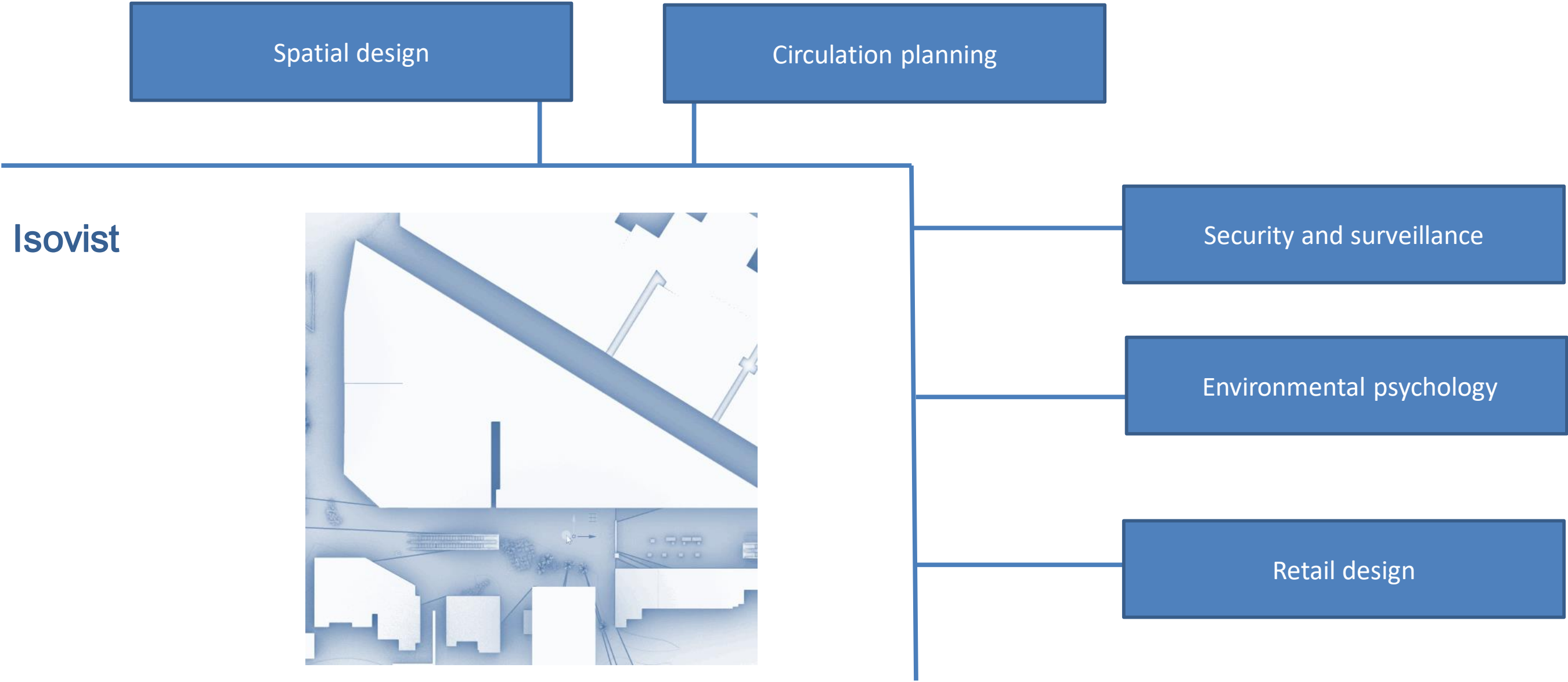
We used it to detect the proper location to create a public place for gathering.

Algorithm usage in architecture and urban planning



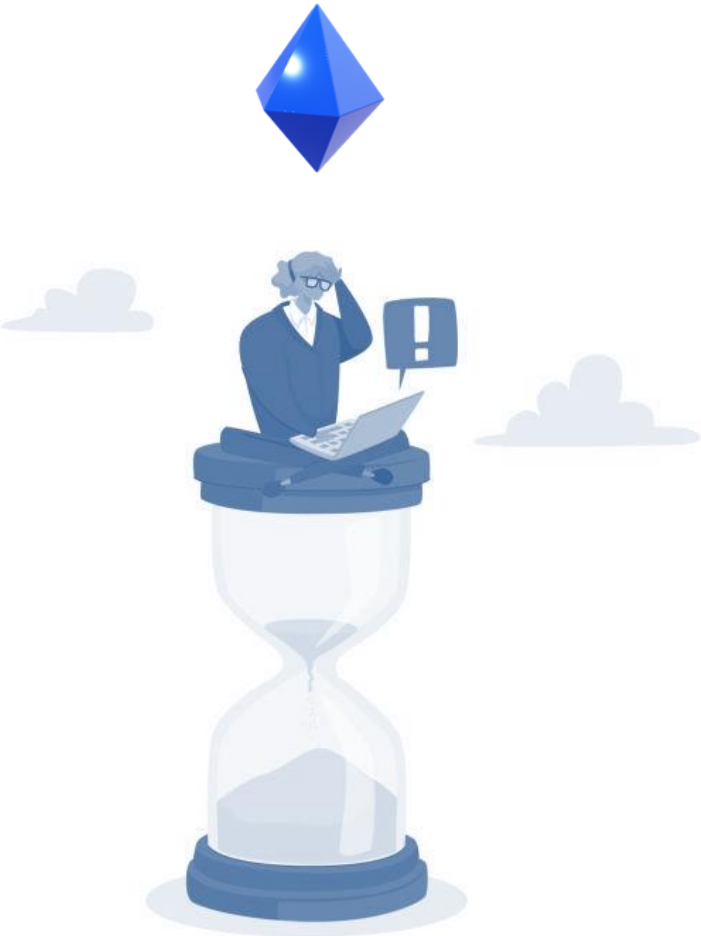
We used it to detect the shortest paths from start to target.

Algorithm usage in architecture and urban planning



We used it to find the place with more openness.

Challenges & Limitations Encountered In The Project



Multiplicity of possibilities considering all criteria

Analysis of the relationship between the mentioned space and the surrounding spaces

The impact of every minor event at the university on this area

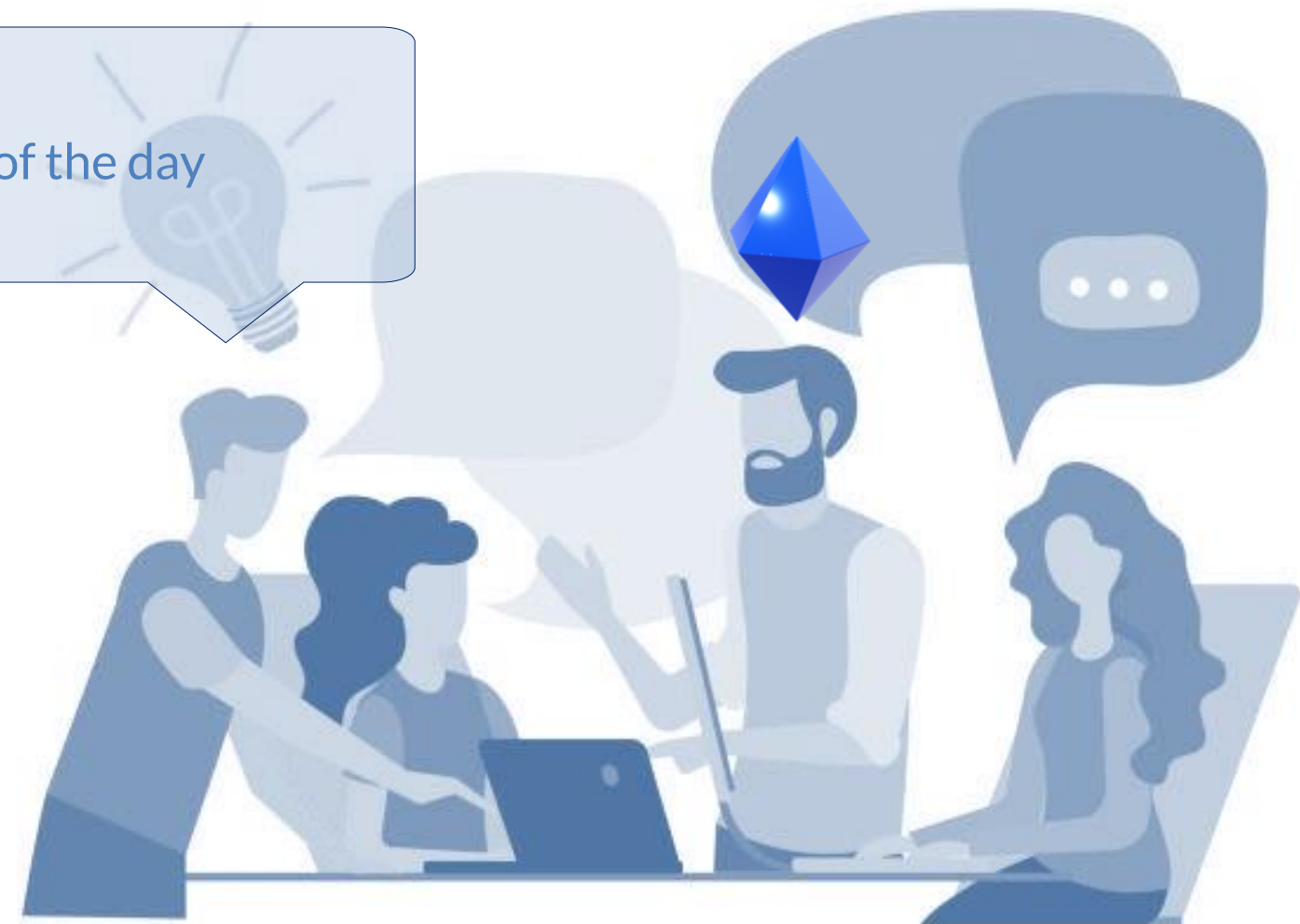
Validating pre-owned models.

Recommendations For Upcoming Researchers

Managing space efficiently based on the crowd

Keeping the incidents in mind and establishing a secure path for users

Be mindful of the varied times of the day



Thank you